ECE 329 **Homework 8** **Due: Mon, Nov 24, 2025, 11:59PM**

5 problems totaling 40pts with 10pts extra Bonus Points. (Bonus problems are optional and the earned bonus credits will be used to "fill in" the "missing points" in the total homework score.)

1. (10pts) For each of the four plane waves (travelling in free space) given below, wave fields $\mathbf{E}$ and $\mathbf{H}$ have the temperal dependence $\propto f(t \mp \frac{s}{c})$ (know as the the d'Alembert wave solution), where s is some coordinate (x, y, or z) and $f(t)$ function is either $\cos(\omega t + \phi)$ or $\sin(\omega t + \phi)$ and $\beta = \omega/c$, a parameter known as wavenumber, and ϕ some "phase shift". Write down the corresponding phasors.

   a) (2.5pts) $\mathbf{E}_1 = 4\sin(\omega t - \beta z)\hat{x}\,\mathrm{V/m}$

   b) (2.5pts) $\mathbf{E}_2 = 5\cos(\omega t - \beta z)\hat{x} - 5\cos(\omega t - \beta z)\hat{y}\,\mathrm{V/m}$

   c) (2.5pts) $\mathbf{H}_3 = 2\cos(\omega t + \beta z + \frac{\pi}{3})\hat{x} - 2\cos(\omega t + \beta z - \frac{\pi}{6})\hat{y}\,\mathrm{A/m}$

   d) (2.5pts) $\mathbf{H}_4 = 4\cos(\omega t + \beta x)\hat{z} + 4\sin(\omega t + \beta x)\hat{y}\,\mathrm{A/m}$.

2. (10pts) Consider an infinite surface current density

$$\mathbf{J_s} = \hat{x} J_{so} \cos(\omega t)$$

flowing on $z = 0$ surface, where $J_{so} > 0$ is real-valued amplitude of the monochromatic surface current measured in A/m units. It is found that $\mathbf{J}_s$ injects field energy into propagating transverse electromagnetic (TEM) waves away from the $z = 0$ plane at an average rate of 8 W/m^2 — that is, the magnitude of the average Poynting vector $\langle \mathbf{S} \rangle$ is 4 W/m^2 for the waves excited above and below the surface current.

a) (3pts) Denoting the TEM waves excited by $\mathbf{J}_s$ (above and below the $z = 0$ plane) as $\mathbf{E} = -\hat{x} E_o \cos(\omega t \mp \beta z)$ V/m and $\mathbf{H} = \mp \hat{y} H_o \cos(\omega t \mp \beta z)$ A/m, where wavenumber $\beta = \frac{\omega}{c}$, determine the numerical values of wave amplitudes E_o and H_o in V/m and A/m units (assuming wave propagation in free space). **Hint:** make use of $\langle \mathbf{S} \rangle$ is 4 W/m^2 information.

b) (3pts) Determine the value of J_{so} in A/m units. **Hint:** consider the boundary condition of $\mathbf{H}$.

c) (4pts) Write down explicitly the phasors of $\mathbf{E}$ and $\mathbf{H}$ fields of part (a) (for regions above and below $z = 0$ plane) using the numerical values of E_o and H_o determined in that part.

3. (10pts) **Damped waves - imperfect dielectric:** A damped plane TEM wave propagating in a material medium is known to have an electric field phasor

$$\tilde{\mathbf{E}} = 4e^{-(0.0001+j0.2)z}\hat{x}\,\frac{\text{V}}{\text{m}}.$$

Clearly, $\gamma = \alpha + j\beta = 0.0001 + j0.2$ 1/m for this wave, with $\alpha \ll \beta$.

When $\alpha \ll \beta$ is true, all the wave parameters η, β, v_p, and λ expressed in terms of μ, ϵ, ω, and f are known to be identical to those for a perfect dielectric except that the attenuation constant α is not 0 but it equals $\frac{\sigma}{2}\sqrt{\frac{\mu}{\epsilon}}$ — you can check the table on page 1 of Prof. Kudeki's Lect 23.

If $\mu = 4\mu_o$ and $v_p = 10^8$ m/s, determine:

a) (1pts) the frequency f of the wave in Hertz (**Hint:** $v_p = \omega/\beta$ and $\omega = 2\pi f$),

b) (1pts) the wavelength λ of the wave (**Hint:** $\beta = 2\pi/\lambda$),

c) (2pts) the dielectric constant $\epsilon_r = \frac{\epsilon}{\epsilon_o}$ of the medium (**Hint:** $\beta \approx \omega\sqrt{\mu\epsilon}$),

d) (2pts) the conductivity σ of the medium (**Hint:** deduce it from α),

e) (2pts) the location z where the peak amplitude of the electric field is $1/e$ of the value at $z = 0$,

f) (2pts) the magnetic field intensity phasor $\tilde{\mathbf{H}}(z)$.

4. (10pts) **Propagation in sea water:** Consider plane TEM wave propagation in a homogeneous material medium with $\mu = \mu_o$, $\epsilon = 81\epsilon_o$, and $\sigma = 4$ S/m appropriate for sea water:

   a) (2pts) At a wave frequency of $f = \frac{\omega}{2\pi} = 10^2$ Hz does the medium behave behave as a "good conductor" or as an "imperfect dielectric"? Explain.

   b) (3pts) Using the approximation formulae appropriate for good conductors or imperfect dielectrics - based on your answer to part (a) - calculate the numerical values of the propagation constant $\gamma \equiv \alpha + j\beta$, wavelength $\lambda = \frac{2\pi}{\beta}$, penetration depth $\delta = \frac{1}{\alpha}$, and intrinsic impedance η for the wave frequency of $f = \frac{\omega}{2\pi} = 10^2$ Hz — use approximations appropriate for the given $\frac{\sigma}{\omega\epsilon}$.

   c) (3pts) Repeat (a) and (b) if $f = \frac{\omega}{2\pi} = 10^9$ Hz.

   d) (2pts) A submarine submerged in the ocean to a depth of 100 m wants to send an electromagnetic signal to a ship a located at the surface. Based on your answers to part (b) and (c), would you recommend the use of $f = \frac{\omega}{2\pi} = 10^2$ Hz or $f = \frac{\omega}{2\pi} = 10^9$ Hz for the transmission signal from the submarine? Discuss.

5. (10pts) **Bonus problem (easy) -Transmission line characteristics:**

a) (5pts) RG-58 is the coax cable that you have been using most frequently in your labs. It has the same geometrical dimensions as the RG-59 cable, but instead of having a dielectric filling with $\epsilon = \epsilon_o$, $\mu = \mu_o$ (like RG-59), it has $\epsilon = 2.25\epsilon_o$, $\mu = \mu_o$. The inner and outer conductor diameters for both types of cables are $2a = 0.032$ inches and $2b = 0.112$ inches, respectively. Given that

$$\mathcal{C} = \epsilon \mathrm{GF}, \quad \mathcal{L} = \frac{\mu}{\mathrm{GF}}, \quad Z_o = \sqrt{\frac{\mathcal{L}}{\mathcal{C}}}, \quad v = \frac{1}{\sqrt{\mathcal{L}\mathcal{C}}},$$

and geometrical factor

$$\mathrm{GF} = \frac{2\pi}{\ln \frac{b}{a}}$$

for a coax. Calculate, $\mathcal{L}$, $\mathcal{C}$, Z_o, v for the RG-58 coax cable.

b) (5pts) $300\,\Omega$ twin-lead transmission lines (TL) are commonly used to connect TV sets and FM radios to their receiving antennas. For the twin-lead, the geometrical factor is

$$\mathrm{GF} = \frac{\pi}{\cosh^{-1} \frac{D}{2a}},$$

where $2a$ is the diameter of each wire (cylindrical conductor) of the twin lead, and D is the distance between the centers of the wires, and $\cosh(\alpha) = \frac{e^{\alpha}+e^{-\alpha}}{2}$ is the hyperbolic cosine function. Assuming $\epsilon = \epsilon_o$, $\mu = \mu_o$, and $a = 1$ mm, calculate D for twin lead TL's having (i) $Z_o = 75\,\Omega$, and (ii) $Z_o = 450\,\Omega$.